



Modelling Financial Markets

Chapter 11: Continuous-Time Models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ CLT: Central Limit Theorem
- ▣ DGS10: 10-Year Treasury Constant Maturity Rate (FRED series code)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JB: Jarque–Bera (test)
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OLS: Ordinary Least Squares
- ▣ OU: Ornstein–Uhlenbeck (process)
- ▣ PDE: Partial Differential Equation
- ▣ QQ: Quantile–Quantile (plot)
- ▣ SDE: Stochastic Differential Equation

Acronyms used in this material (2/2)

- ▣ **SVJ: Stochastic Volatility with Jumps (model)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**
- ▣ **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Outline

Motivation

From Random Walk to Brownian Motion

Quadratic Variation and Itô Calculus

Stochastic Differential Equations and Simulation

Geometric Brownian Motion

Mean Reversion: the Ornstein–Uhlenbeck Process

Estimating Diffusions: What Is Identified from What

Change of Measure: Girsanov and Feynman–Kac

Jumps: the Merton Model

Stochastic Volatility: the Heston Model

Models against the Data

AI for Scientific Discovery

Summary

References

Why Continuous Time?

The question of this chapter

- Prices are recorded daily or every few milliseconds, but trading never really stops
- Can we describe prices with a model that works at **every** time scale at once?
- Option pricing (Chapter 12), interest-rate models and risk management are written in this language

What we do in this chapter

- Build Brownian motion and Itô calculus through simulation
- Simulate stochastic differential equations and measure the error of the schemes
- Estimate four models on data: geometric Brownian motion, Ornstein–Uhlenbeck, Merton jumps, Heston volatility
- Confront every model with the stylised facts of Chapter 1

From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
 - ▶ The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC
BY-SA 2.0 de; Wikimedia Commons



Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd
(1970–1975); public domain;
Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

Data Used in This Chapter

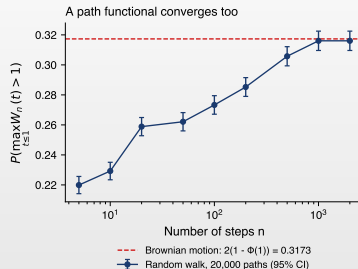
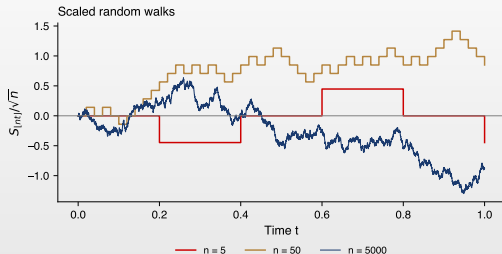
Series	Type	Period	Days
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$
- Each series on its own calendar; equity indices on weekdays, stale quotes removed (unchanged close and no intraday range); genuine zero returns kept
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: 1/365); drifts and volatilities annualised
- W_t : standard Brownian motion; S_t : price; v_t : instantaneous variance; $\mathcal{F}_t$: the information (filtration) up to t

From Random Walk to Brownian Motion: Donsker's Theorem

- Scaled random walk $W_n(t) = S_{\lfloor nt \rfloor} / \sqrt{n}$, $S_k = \xi_1 + \dots + \xi_k$, steps i.i.d. with mean 0 and variance 1
 - ▶ Time step $1/n$, space step $1/\sqrt{n}$: the only scaling with a non-trivial limit; by the central limit theorem (CLT), $W_n(t) \rightarrow N(0, t)$ for each t
- Donsker: $W_n \Rightarrow W$ as random functions, i.e. for the **whole path** Billingsley (1999)
 - ▶ “Invariance”: the limit does not depend on the distribution of the steps
- Continuous mapping: every continuous functional of the path converges too
 - ▶ Example: the maximum; by the reflection principle $P(\max_{t \leq 1} W_t > a) = 2P(W_1 > a)$
- For finance: many small independent trades add up to Brownian-like price paths

Random Walks Converge to Brownian Motion

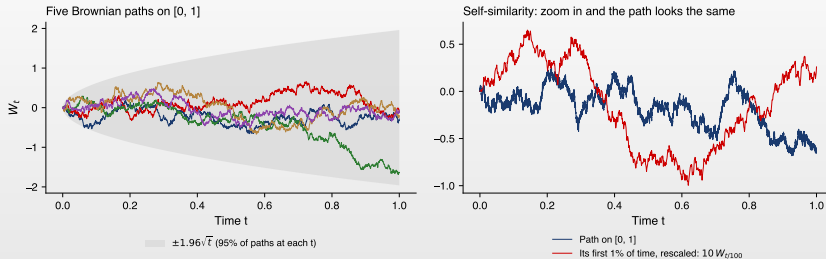


- Left: one scaled random walk for $n = 5, 50, 5000$ steps; right: probability that the path exceeds 1 before $t = 1$
- $P(\max W_n > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), against the Brownian $2(1 - \Phi(1)) = 0.3173$ (Φ : standard Normal distribution function)
- A coarse grid misses the maximum between observations, and convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range

Brownian Motion: Definition and Properties

- **Standard Brownian motion:** $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t - s)$ for $s < t$; continuous paths
 - ▶ Existence: [Wiener \(1923\)](#); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t} Z$, $Z \sim N(0, 1)$
- $\text{Cov}(W_s, W_t) = \min(s, t)$; a Gaussian process and a **martingale**: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ (the efficient-market random walk of Chapter 2)
- **Self-similarity:** $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
 - ▶ $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \rightarrow \infty$: ordinary calculus fails

Brownian Paths

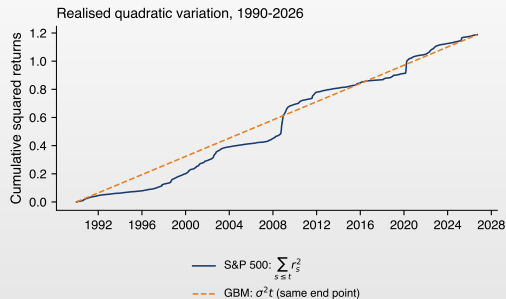
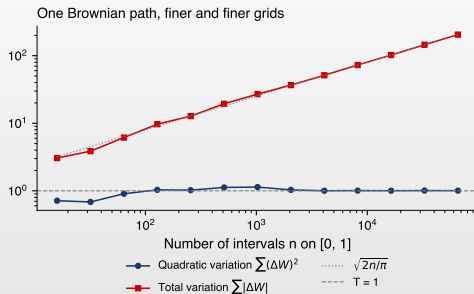


- Left: five paths with the band $\pm 1.96\sqrt{t}$; right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of Chapter 7); a zoomed piece looks like the whole path
- Real markets break self-similarity: fat tails and clustering depend on the horizon (Chapter 1)

Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \dots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
 - ▶ Its mean is $\sqrt{2nT/\pi} \rightarrow \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
 - ▶ A random sum with a **deterministic** limit
 - ▶ $\text{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance (Chapter 9)
- Shorthand used in Itô calculus: $(dW_t)^2 = dt$

Quadratic Variation: Simulation and Data



□ Left: one Brownian path on grids with 16 to 65,536 intervals; right: cumulative sum of squared daily S&P 500 log returns

 MFM_ch11_quadratic_variation

Reading the Quadratic Variation Chart

- ▣ Simulation: the quadratic variation settles at 1.006 ($T = 1$); the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- ▣ S&P 500: total $\sum r_t^2 = 1.188$ over 36.7 years, an average volatility of 18.0%
 - ▶ Under geometric Brownian motion the curve would be a straight line
- ▣ It moves in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of all quadratic variation
 - ▶ February – April 2020: 0.5% of the days, 7.8% of the quadratic variation
- ▣ Volatility is itself random: the motivation for the Heston model at the end of the chapter

The Itô Integral

- Goal: give a meaning to $\int_0^T H_t dW_t$, the gain of holding H_t units of an asset whose price moves like W
 - ▶ Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit of sums with the integrand at the **left** end point
 - ▶ $\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})$ (limit in mean square)
- H must be **adapted** (progressively measurable) and square-integrable, $\mathbb{E} \left[\int_0^T H_t^2 dt \right] < \infty$
 - ▶ Financial reading: H_t uses only information up to t ; you choose the position before you see the next price move
 - ▶ Left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes

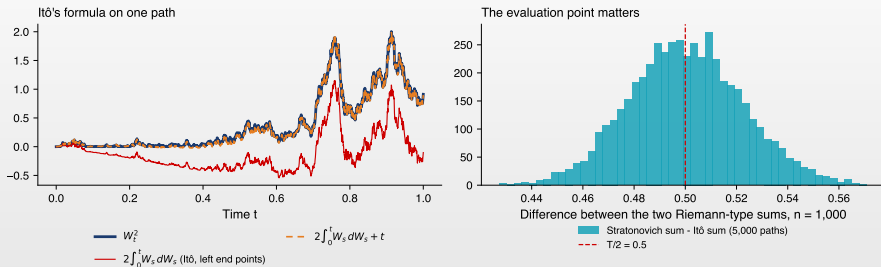
Properties of the Itô Integral

- For square-integrable adapted H (previous slide):
- Zero mean: $\mathbb{E} \left[\int_0^T H_t dW_t \right] = 0$: no strategy earns money from a martingale on average
- **Itô isometry**: $\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$
- The process $t \mapsto \int_0^t H_s dW_s$ is a martingale
 - ▶ If only $\int_0^T H_t^2 dt < \infty$ almost surely, it is merely a local martingale, and its mean need not be zero
- The evaluation point matters: the mid-point rule gives the **Stratonovich** integral, a different number
 - ▶ Finance uses Itô, because a mid-point position would look into the future

Worked Example: $\int_0^T W_t dW_t$

- ▣ Identity for each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2} (W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2} (\Delta W_k)^2$
- ▣ Sum over k : the first part telescopes to $\frac{1}{2} W_T^2$, the second tends to $\frac{1}{2} T$
 - ▶ $\int_0^T W_t dW_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
- ▣ Ordinary calculus would give $\frac{1}{2} W_T^2$; the extra $-\frac{1}{2} T$ is the quadratic variation
- ▣ Check: the mean is $\frac{1}{2}(T - T) = 0$, as it must be for an Itô integral

Itô's Formula on Simulated Paths



- Left: W_t^2 , the Itô integral $2 \int_0^t W \, dW$ and their difference t on one path; right: Stratonovich minus Itô sums over 5,000 paths
- $2 \int_0^t W \, dW + t$ lies exactly on W_t^2 : the formula holds path by path; Stratonovich minus Itô: mean 0.500, s.d. 0.022 (theory $T/2 = 0.5$)
- Itô sums: mean 0.007, variance 0.520; theory: 0 and, by the isometry, $\int_0^1 t \, dt = 1/2$

Itô Processes and Itô's Lemma

- **Itô process:** $dX_t = a_t dt + b_t dW_t$ (drift a_t , diffusion coefficient b_t , both adapted)
 - ▶ Shorthand for $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$
- **Itô's lemma:** for $f(t, x)$ twice differentiable [Itô \(1944\)](#)
 - ▶ $df(t, X_t) = (f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx}) dt + b_t f_x dW_t$
- Rule of thumb: Taylor expansion to second order with $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$
 - ▶ The second-order term $\frac{1}{2} b_t^2 f_{xx} dt$ does not vanish: this is the whole novelty

Worked Example: the Logarithm of a Price

- Price dynamics $dS_t = \mu S_t dt + \sigma S_t dW_t$; take $f(x) = \ln x$, so $f_x = 1/x$, $f_{xx} = -1/x^2$
- Itô's lemma with $a_t = \mu S_t$, $b_t = \sigma S_t$
 - ▶ $d \ln S_t = \left(\mu S_t \cdot \frac{1}{S_t} - \frac{1}{2} \sigma^2 S_t^2 \cdot \frac{1}{S_t^2} \right) dt + \sigma S_t \cdot \frac{1}{S_t} dW_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$
- The log price grows at $\mu - \sigma^2/2$, not at μ : volatility lowers the growth rate of the median investor's wealth
- With $\sigma = 18\%$: $\sigma^2/2 = 1.6$ percentage points per year; with $\sigma = 67\%$ (Bitcoin): 22 points

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \Delta Y_k$; if $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$, then $d[W^{(1)}, W^{(2)}]_t = \rho dt$
 - ▶ Product rule: $d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ twice differentiable
 - ▶ $df(t, X_t) = f_t dt + \sum_i f_{x_i} dX_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} d[X^i, X^j]_t$
- Worked example (Heston, Section 10): $dS = \mu S dt + \sqrt{v} S dW^{(1)}$, $dv = \kappa(\theta - v) dt + \xi \sqrt{v} dW^{(2)}$
 - ▶ $d(S_t v_t) = S_t dv_t + v_t dS_t + \rho \xi v_t S_t dt$
 - ▶ ρ enters the drift of the product: with $\rho < 0$, price and variance tend to move in opposite directions

Recap: Itô Calculus

- Brownian paths have infinite length but finite quadratic variation $[W]_t = t$
- The Itô integral uses left end points: for square-integrable integrands it is a martingale with zero mean
- Itô's lemma = Taylor to second order with $(dW)^2 = dt$; with several Brownian motions add $d[W^{(1)}, W^{(2)}] = \rho dt$
- Real prices have a quadratic variation that grows in bursts, not linearly

Stochastic Differential Equations

- A **stochastic differential equation** (SDE): $dX_t = a(t, X_t) dt + b(t, X_t) dW_t$, X_0 given
 - ▶ Drift a and diffusion b now depend on the current state
- Existence and uniqueness of a solution: a and b Lipschitz in x with at most linear growth [Kloeden & Platen \(1992\)](#)
- Few SDEs have closed-form solutions; the four models of this chapter are among them
 - ▶ Geometric Brownian motion, Ornstein–Uhlenbeck and Merton: exact; Heston: only the characteristic function
- Otherwise we simulate on a grid; the question is how large the discretisation error is

Euler–Maruyama and Milstein Schemes

- **Euler–Maruyama:** freeze a and b over each step [Higham \(2001\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k, \Delta W_k = \sqrt{\Delta t} Z_k$

- **Milstein:** add the next Itô–Taylor term [Milstein \(1975\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k + \frac{1}{2} b(X_k) b'(X_k) [(\Delta W_k)^2 - \Delta t]$

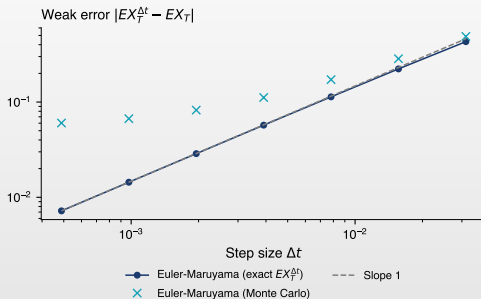
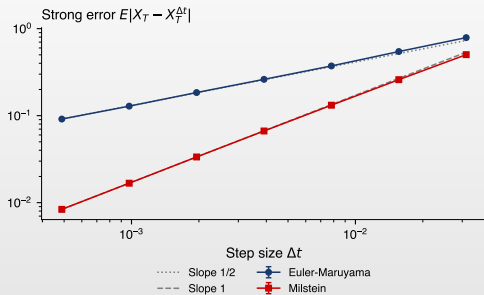
- The correction uses $(\Delta W_k)^2 - \Delta t$, which has mean zero: it only matters path by path

- Test equation [Higham \(2001\)](#): $dX = \lambda X dt + \mu X dW$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark

Strong and Weak Convergence

- **Strong error:** $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C \Delta t^\gamma$, how close each simulated path is to the true one
 - ▶ Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$
 - ▶ Needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error:** $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C \Delta t^\beta$, how close the distribution is
 - ▶ Both schemes: $\beta = 1$; enough for prices of European options, VaR, expected values
- On a log-log plot, the order is the slope of error against step size

Convergence of the Schemes



□ 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5} ; bars are 95% CIs

 MFM_ch11_convergence

Reading the Convergence Chart

- ▣ Estimated strong orders (slope of the log-log fit; 95% CI by resampling whole paths, all step sizes together): Euler–Maruyama 0.518 [0.512, 0.525], Milstein 0.986 [0.984, 0.987]
 - ▶ Close to the theoretical $1/2$ and 1 ; the narrow intervals measure Monte Carlo noise only, and the small gap is finite-step curvature, not a different order
 - ▶ Halving Δt cuts the Milstein error in half, the Euler error only by 29%
- ▣ Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope with no sampling error
 - ▶ The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias
- ▣ In practice the sampling error of Monte Carlo, not the step size, is often the binding constraint [Glasserman \(2003\)](#)

Recap: Simulation

- An SDE specifies the local drift and diffusion of a process
- Euler–Maruyama is simple and strong order $1/2$; Milstein adds one term and reaches order 1
- For expected values the weak order 1 is what matters, together with the Monte Carlo sample size
- When the exact transition is known (next sections), simulate exactly: no discretisation error

Geometric Brownian Motion (GBM)

- SDE: $dS_t = \mu S_t dt + \sigma S_t dW_t$, i.e. the **percentage** change has constant drift and volatility
 - ▶ μ : expected rate of return; σ : volatility, both per year
- Solution (from the log example): $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right)$
 - ▶ Prices stay positive; exact simulation needs only Normal draws
- The price model behind [Black & Scholes \(1973\)](#); proposed by [Osborne \(1959\)](#) and [Samuelson \(1965\)](#)
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), chapters on the Wiener process and stochastic integration

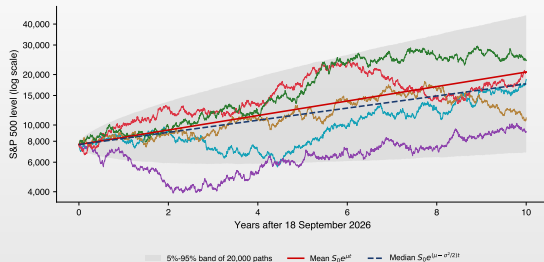
Log Returns under GBM

- Log returns over Δt : $r_t = \ln(S_t/S_{t-\Delta t}) \sim N((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2\Delta t)$
 - ▶ Independent and identically distributed (i.i.d.) at every frequency
- Testable implications
 - ▶ Normal log returns: skewness 0, excess kurtosis 0
 - ▶ No autocorrelation in r_t , $|r_t|$ or r_t^2
 - ▶ Variance proportional to the horizon
- S_T/S_0 is **lognormal**: $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with the log drift $m = \mu - \sigma^2/2$

Mean, Median and Mode of a Lognormal Price

- Mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$; median $S_0 e^{mT}$; mode $S_0 e^{(m-\sigma^2)T}$: mode < median < mean
- Worked example, S&P 500 parameters: $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$, $T = 10$ years
 - ▶ Mean gross return 2.71, median 2.30, mode 1.66
 - ▶ $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
 - ▶ $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end **below** the average
- The average is pulled up by a few very large outcomes; the gap grows with $\sigma^2 T$ (Bitcoin: $\sigma \approx 67\%$)
- For long horizons, report the median and quantiles, not only the mean

Ten Years of Simulated S&P 500 Paths



- GBM with $\hat{\mu}$, $\hat{\sigma}$ from 1990–2026, starting at the close of 18 September 2026 ($S_0 = 7\,650$); 20,000 exact paths, five shown
- After ten years: mean 20 696, median 17 604, 5%–95% range 6 878 – 45 017; $P(S_T < S_0) = 7.1\%$
- Too narrow in crises and too smooth: GBM has no fat tails and no volatility clustering (tested below)

Estimating GBM by Maximum Likelihood

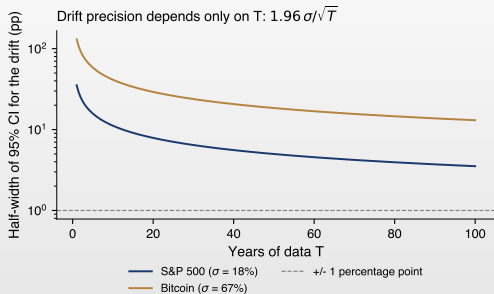
- Maximum likelihood estimation (MLE) from n log returns at step Δt , over $T = n\Delta t$ years
 - ▶ $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
- Standard errors under GBM (i.i.d. Normal returns): $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ and $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$
 - ▶ Volatility precision improves with more observations (sample more often)
 - ▶ Drift precision depends only on the **calendar length** T : daily or minute data do not help
- Reason: $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

Worked Example: GBM for Three Markets

	Years	$\hat{\sigma}$ (%)	s.e.($\hat{\sigma}$)	$\hat{m}$ (%)	s.e.($\hat{m}$)
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

- Under GBM, volatility is pinned down to about 0.1–0.7 percentage points; the log drift only to 3–19 points
- These s.e. are conditional on GBM: with excess kurtosis k , even independent returns inflate s.e.($\hat{\sigma}$) by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap s.e. in Seminar B2
- S&P 500: $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2 = 8.3\% + 1.62\% = 10.0\%$ per year
- A 95% CI of ± 1 point for the S&P 500 drift would need $(1.96 \hat{\sigma}/0.01)^2 = 1243$ years of data

How Precisely Can We Estimate the Drift?



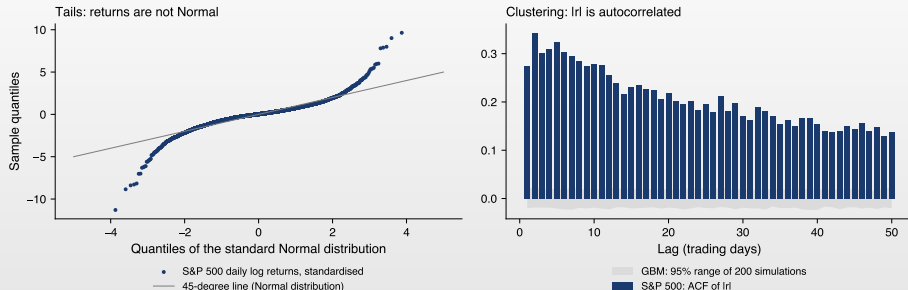
□ Left: half-width of the 95% CI for the drift against years of data; right: S&P 500 estimates on rolling 10-year windows

 MFM_ch11_drift_precision

Reading the Drift Chart

- Ten-year log drift of the S&P 500 ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09)
 - ▶ Every window's 95% CI is about ± 11 points wide: the variation is mostly noise
- Ten-year volatility stays between 14.1% and 22.3% : estimated precisely, but not constant
- Bitcoin would need 17 028 years for ± 1 point: its expected return is essentially unknowable from its price history
- Option pricing in Chapter 12 avoids μ : under the risk-neutral measure only σ enters (Girsanov, Section 8)

Does GBM Fit the S&P 500?



- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution; right: autocorrelation function (ACF) of $|r_t|$ against 200 GBM samples of the same length

 MFM_ch11_gbm_vs_data

Reading the GBM Test

- Tails: excess kurtosis 10.9 instead of 0; the worst day is -11.3 standard deviations
 - ▶ 62 days beyond ± 4 standard deviations, against 0.59 expected under GBM
 - ▶ Jarque–Bera (JB) statistic 45 805: Normality is rejected at any level
- Clustering: ACF of $|r_t|$ is 0.27 at lag 1 and still 0.14 at lag 50
 - ▶ GBM samples never exceed 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 [Cont \(2001\)](#); the rest of the chapter adds mean reversion, jumps and random volatility

Recap: Geometric Brownian Motion

- GBM: constant percentage drift and volatility; lognormal prices; i.i.d. Normal log returns
- The median grows at $\mu - \sigma^2/2$; the mean at μ ; most paths end below the mean
- Volatility is estimable from data; the drift is not, in any realistic sample
- The data reject GBM: fat tails and volatility clustering

The Ornstein–Uhlenbeck (OU) Process

- SDE: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930)
 - ▶ θ : long-run mean; κ : speed of mean reversion; σ : volatility
 - ▶ Above θ the drift is negative, below θ positive: a spring pulling x back
- Used for interest rates, volatility, spreads, commodity prices, exchange-rate deviations
 - ▶ Vasicek (1977): the OU process as a model of the short-term interest rate

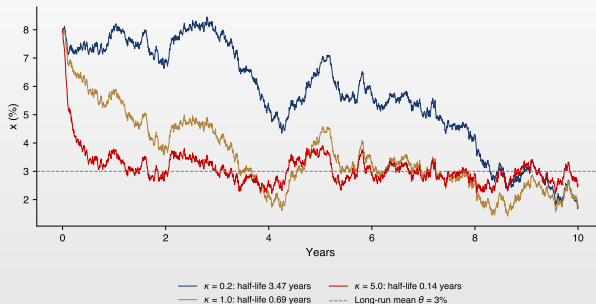
Solution and Stationary Distribution

- By Itô's lemma on $e^{\kappa t} x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$
- Conditional distribution: Normal with
 - ▶ $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t})$
- Stationary distribution ($t \rightarrow \infty$): $N(\theta, \sigma^2/(2\kappa))$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
 - ▶ $\kappa = 0.5$ per year: half-life 1.39 years

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
 - ▶ $x_{t+\Delta t} = a + b x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1 - b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t , then invert the mapping
 - ▶ $\hat{\kappa} = -\ln \hat{b} / \Delta t$, $\hat{\theta} = \hat{a} / (1 - \hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa} / (1 - \hat{b}^2)}$
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion



- OU paths driven by the same shocks, starting at 8%, with $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps

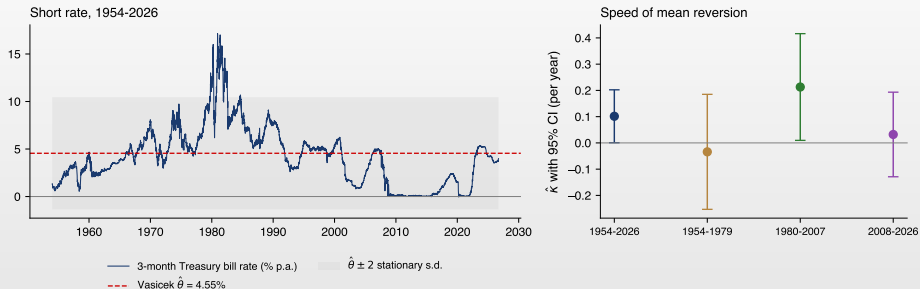
Reading the OU Chart

- Half-lives: 3.47, 0.69 and 0.14 years
 - ▶ With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%
 - ▶ Fast reversion keeps the process in a narrow band; slow reversion looks like a random walk for years
- Unlike Brownian motion, the variance stops growing: long-horizon forecasts converge to θ

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process [Vasicek \(1977\)](#)
 - ▶ Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
 - ▶ Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross (CIR)**: $dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t$ [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) [Feller \(1951\)](#)
- Which is closer to the data is an empirical question, tested in Section 7 [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)

Vasicek on the 3-Month Treasury Bill Rate



- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean; right: $\hat{\kappa}$ with 95% CI by period

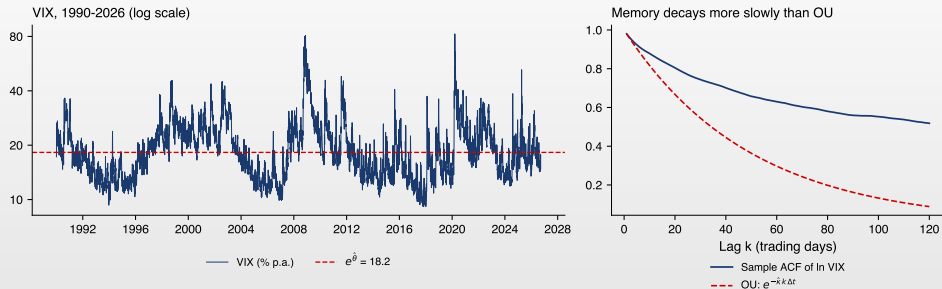
Reading the Vasicek Chart

- ▣ Full sample: $\hat{\theta} = 4.55\%$, $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$
 - ▶ Daily AR(1) coefficient $\hat{b} = 0.99960$: almost a unit root
- ▣ Subperiods disagree: $\hat{\kappa} = -0.034$ (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026)
 - ▶ Two of the three 95% CIs include zero: mean reversion is not detectable within 20–25 years
- ▣ The rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of days were below 0.05%
 - ▶ The fitted Vasicek model gives a 6.1% long-run probability of a negative rate

The Mean-Reversion Speed Is Biased Upwards

- In finite samples $\hat{\kappa}$ overstates κ ; the bias is roughly $4/T$ with T in years [Tang & Chen \(2009\)](#)
 - ▶ It depends on the calendar span, not on the sampling frequency, like the drift of GBM
- Simulation with the full-sample Vasicek estimates as the truth (72 years, 500 samples)
 - ▶ Mean $\hat{\kappa} = 0.164$ against the true 0.101: bias 0.063, close to $4/T = 0.055$
 - ▶ 78% of the estimates exceed the truth
- Consequence: estimated half-lives are too short, and mean reversion looks faster than it is

Is Volatility Mean-Reverting? The VIX



- Left: VIX 1990–2026 on a log scale with the OU long-run level of $\ln VIX$; right: sample ACF of $\ln VIX$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$

Reading the VIX Chart

- OU fit to $\ln VIX$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days, long-run level $e^{\hat{\theta}} = 18.2$
 - ▶ Volatility spikes (VIX 82.7 on 2020-03-16) fade within weeks, not years
- But the ACF at lag 60 is 0.63, against 0.30 implied by OU
 - ▶ Fast decay first, then a slow component: one OU factor is too simple
 - ▶ Candidates: two-factor or long-memory models; rough volatility [Gatheral, Jaisson & Rosenbaum \(2018\)](#) is a distinct alternative without long memory; which fits better must be tested
- Mean-reverting volatility is the building block of the Heston model

Recap: Mean Reversion

- OU: a linear pull towards θ with speed κ ; Normal, stationary, exactly an AR(1) when sampled
- Half-life $\ln 2 / \kappa$ summarises the persistence
- Interest rates: κ small, unstable across periods and biased upwards in finite samples
- Volatility (VIX): clear mean reversion within weeks, plus a slow component

Infill versus Long-Span Asymptotics

- **Infill:** $\Delta \rightarrow 0$ on a fixed $[0, T]$; then $\sum_k (\Delta X_k)^2 \rightarrow \int_0^T b_t^2 dt$
 - ▶ The diffusion coefficient is identified **without error** in the limit: quadratic variation is observable
- The drift is **not** consistently estimable on a fixed span: its Fisher information grows with T , not with n
 - ▶ GBM: $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ (Section 5); OU: $\text{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \rightarrow 2\kappa/T$ as $\Delta \rightarrow 0$ (Seminar A9)
- **Long span** ($T \rightarrow \infty$) is needed for μ, κ, θ ; the bias of $\hat{\kappa}$ is not reduced by finer sampling [Yu \(2012\)](#)
 - ▶ $b = e^{-\kappa\Delta} \rightarrow 1$: a near unit root, so $\hat{\kappa}$ is far from Normal when κT is small
- One phenomenon, not two facts: the imprecise GBM drift and the biased $\hat{\kappa}$ are both long-span problems

Likelihood for Nonlinear Diffusions

- **Euler pseudo-likelihood:** a Gaussian transition $r_{t+\Delta} \mid r_t \sim N(r_t + a(r_t)\Delta, b^2(r_t)\Delta)$
 - ▶ Biased at order Δ : small with daily data, larger with monthly or quarterly steps
- **Exact CIR:** $2c r_{t+\Delta} \mid r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ $c = 2\kappa/(\sigma^2(1 - e^{-\kappa\Delta}))$, noncentrality $2c r_t e^{-\kappa\Delta}$; Feller condition $\Leftrightarrow$ degrees of freedom ≥ 2 (Seminar A5)
- General diffusions: closed-form Hermite expansions of the transition density [Aït-Sahalia \(2002\)](#)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)
 - ▶ Gaussian (Euler) quasi-likelihood; tests on γ with sandwich standard errors whose score covariance is heteroskedasticity- and autocorrelation-consistent (HAC) [Newey & West \(1987\)](#)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007

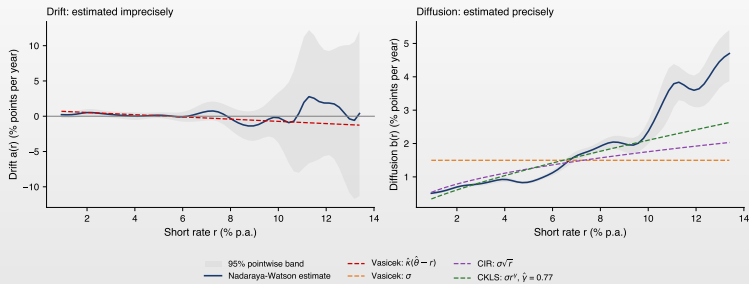
Model	$\hat{\kappa}$ daily (s.e.)	$\hat{\kappa}$ monthly (s.e.)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033); 0.60 (0.149)

- Daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$); 2008–2026 excluded: 13 exact zeros and 8 negative quotes lie outside the domain of these likelihoods, and 17% of the days are a near-zero regime below 0.05%
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- HAC Wald tests (11 and 6 lags): $\gamma = 0$ rejected (daily 545, monthly 16); $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
- The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) found $\hat{\gamma} \approx 1.5$ for 1964–1989

Nonparametric Drift and Diffusion

- No functional form: kernel averages of the increments around each level x
 - ▶ $\hat{a}(x) = \sum_t w_t(x) \Delta r_t / \Delta$, $\hat{b}^2(x) = \sum_t w_t(x) (\Delta r_t)^2 / \Delta$, $w_t(x) \propto K((r_t - x)/h)$
 - ▶ Nadaraya–Watson weights, Gaussian kernel K , bandwidth h ; $\mathbb{E}[\Delta r \mid r = x] / \Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x] / \Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \rightarrow 0$
- Stanton (1997): nonparametric drift, diffusion and market price of interest-rate risk
- Bandi & Phillips (2003): valid for non-stationary but recurrent processes, which matters because the T-bill rate is close to a unit root
- The diffusion converges at the infill rate, the drift only at the long-span rate
- Caution: the nonparametric specification test of Aït-Sahalia (1996) rejects far too often with persistent data Pritsker (1998)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Daily DTB3, 1954–2007; bandwidth $h = 0.44$ percentage points; 95% heteroskedasticity-robust pointwise bands (serial dependence ignored)
- Drift: zero lies inside the band at 97% of the levels; diffusion: median relative s.e. 8%, $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits everywhere: inside the band at 35% (CKLS), 7% (CIR), 5% (Vasicek) of the levels

Recap: Estimating Diffusions

- Infill identifies the diffusion coefficient; the drift, κ and θ need a long calendar span
- Euler versus exact likelihood: the discretisation bias is small next to the sampling error of the drift
- Short-rate volatility rises with the level: Vasicek is rejected; γ lies between 1/2 and 1 depending on the frequency
- Nonparametric estimates: precise diffusion, imprecise drift; specification tests on persistent data over-reject

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by a **Radon–Nikodym density**
 - ▶ $\frac{dQ}{dP} \Big|_{\mathcal{F}_T} = Z_T = \exp \left(- \int_0^T \theta_s dW_s - \frac{1}{2} \int_0^T \theta_s^2 ds \right)$
- **Girsanov (1960)**: under Q , $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a Brownian motion if $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \theta_s^2 ds} \right] < \infty$ (Novikov condition)
- For $dX = a dt + b dW$: $dX = (a - b\theta) dt + b d\tilde{W}$
 - ▶ **Only the drift changes**: the diffusion coefficient and the quadratic variation are the same under P and Q
 - ▶ Under GBM (constant σ), realised variance (Chapter 9) therefore measures the σ that prices options, while μ drops out
 - ▶ With stochastic volatility, P and Q weigh future variance paths differently: historical variance is a benchmark, not a calibration (next slide)
- Textbook treatment: [Shreve \(2004\)](#), Chapters 5–6; [Franke, Härdle & Hafner \(2019\)](#)

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q , $dS = rS dt + \sigma S d\tilde{W}$ (Seminar A1)
 - ▶ S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, s.e. $1/\sqrt{T} = 0.17$ (r : mean 3-month T-bill rate)
 - ▶ The discounted price $e^{-rt}S_t$ is a Q -martingale
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$
 - ▶ The long-run level that prices bonds differs from the one estimated on data (Section 6)
- Heston with volatility risk premium λv_t : $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
 - ▶ $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE

- If $dX = a(X) dt + b(X) d\tilde{W}$ under Q Kac (1949)
 - ▶ $u(t, x) = \mathbb{E}^Q \left[e^{-\int_t^T r_s ds} g(X_T) \mid X_t = x \right]$ solves $u_t + a u_x + \frac{1}{2} b^2 u_{xx} - r u = 0$, $u(T, x) = g(x)$
- Proof idea: $e^{-\int_0^t r_s ds} u(t, X_t)$ is a Q -martingale, so Itô's lemma must give it a zero drift
- Vasicek bond ($g = 1$, $X = r$): $P(t, T) = \exp(A(\tau) - B(\tau) r_t)$, $\tau = T - t$ (Seminar A3)
 - ▶ $B(\tau) = \frac{1 - e^{-\kappa \tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2} \right) (B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
 - ▶ Section 6 estimates with $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05%; the 10-year Treasury yield (FRED DGS10) on 2026-09-18: 5.01%
- The gap is a term premium ($\lambda \neq 0$) plus misspecification; Chapter 12 derives Black–Scholes by the same route

Recap: Change of Measure

- Girsanov: an equivalent change of measure shifts the drift by $-b\theta$ and leaves the volatility unchanged
- The market price of risk links P and Q : drift parameters differ across measures, diffusion parameters do not
- Feynman–Kac turns a Q -expectation into a pricing PDE: bond prices here, option prices in Chapter 12
- The market price of risk for the S&P 500 is estimated with a standard error of 0.17 from 37 years: as imprecise as the drift

Why Jumps? Poisson and Compound Poisson



Amsterdam Stock Exchange, 21 October 1987, after
the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia
Commons

- 19 October 1987: the Dow Jones fell 22.6% in one day, about 20 standard deviations under GBM with 18% volatility
- Brownian paths are continuous; real prices gap on news, and a jump cannot be hedged by continuous trading [Merton \(1976\)](#)
- N_t Poisson with intensity λ ; **compound Poisson** $J_t = \sum_{i \leq N_t} Y_i$, sizes Y_i i.i.d., independent of N
 - Cumulants $\kappa_k(J_t) = \lambda t \mathbb{E}[Y^k]$; Merton: $Y_i \sim N(\mu_J, \sigma_J^2)$, jumps in the **log** price

Merton's Jump-Diffusion

- Log price = Brownian motion with drift + compound Poisson jumps [Merton \(1976\)](#)
 - ▶ $d \ln S_t = m dt + \sigma dW_t + dJ_t$, $J_t = \sum_{i \leq N_t} Y_i$
- Return over Δt given k jumps: $N(m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
 - ▶ Unconditional density: a **Poisson mixture** of Normal densities
 - ▶ $f(r) = \sum_{k \geq 0} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!} \varphi(r; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
- Five parameters ($m, \sigma, \lambda, \mu_J, \sigma_J$); returns still i.i.d. over time
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then a Normal with the conditional moments

Moments of the Merton Return

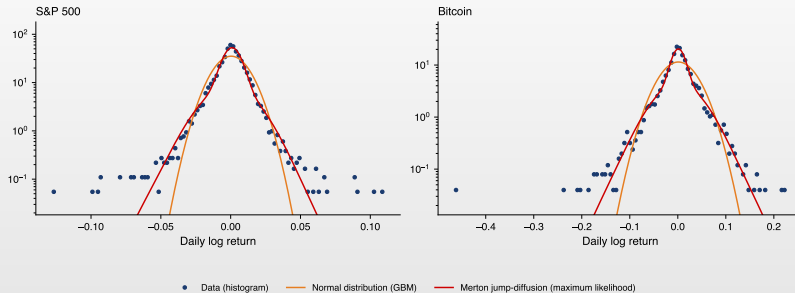
- Cumulants of the return over Δt (the k -th cumulant of a compound Poisson sum is $\lambda \Delta t \mathbb{E}[Y^k]$):
 - ▶ $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
 - ▶ $\kappa_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $\kappa_3 / \kappa_2^{3/2}$, excess kurtosis κ_4 / κ_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
 - ▶ Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07

Maximum Likelihood Estimates

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	-0.16	1.44	71%	2 401
BET	11.3	79 (7)	-0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day), standard errors from the Hessian in brackets
- LR against GBM ($\lambda = 0$) is not χ^2_3 : $\lambda = 0$ is on the boundary and μ_J, σ_J are unidentified; bootstrap critical values in Seminar B6
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution



- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture

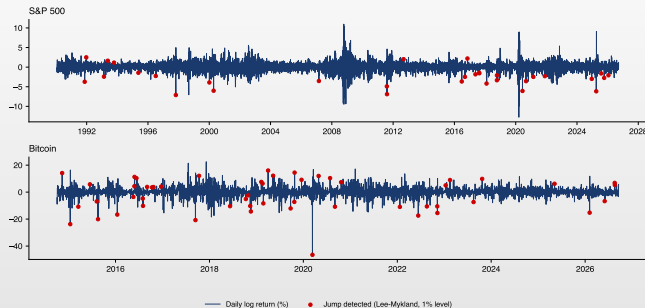
Reading the Merton Fit

- The Merton density follows the peak and the shoulders far better than the Normal distribution
 - ▶ LR statistics in the thousands, far beyond the bootstrap critical value: GBM is rejected
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data; skewness -0.31 against -0.36
 - ▶ The most extreme days lie beyond both fitted curves
- About 102 “jumps” per year means the model uses jumps to mimic **changing volatility**
 - ▶ An i.i.d. model cannot tell a jump from a volatile period: we need a direct test for jumps
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \rightarrow 0$, that density explodes; the MLE is a local maximum, and the LR test is defined on $\sigma \geq 5\%$ (Seminar B6) [Honoré \(1998\)](#)

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
 - ▶ $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
 - ▶ Reject if $(|L_t| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$; at $\alpha = 1\%$ this threshold for the **normalised** statistic, not for $|L_t|$, is 4.60
 - ▶ $C_n = \frac{\sqrt{2 \ln n}}{c} - \frac{\ln \pi + \ln \ln n}{2c\sqrt{2 \ln n}}$, $S_n = \frac{1}{c\sqrt{2 \ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin



□ Daily log returns (%) with the days flagged by the Lee–Mykland test at the 1% level

Reading the Jump Chart

- ▣ S&P 500: 32 jumps in 36.7 years, 0.87 per year; 84% of them are falls
 - ▶ Not one of the 10 largest absolute returns is flagged: they came in turbulent periods (2008, 2020)
 - ▶ Instead, moves as small as 1.2% in a very calm market are jumps (1993-10-14)
- ▣ Bitcoin: 52 jumps, 4.35 per year, 5 of its 10 largest moves flagged
 - ▶ Up and down jumps equally often (52% negative)
- ▣ The test flags only a few candidate jumps per year, far fewer than the Merton fit; the gap may reflect stochastic volatility, drift bursts and limited detection power in daily data

Recap: Jumps

- Merton adds compound Poisson jumps to GBM; returns become a Poisson mixture of Normal distributions
- Jumps create skewness and excess kurtosis, strongest at short horizons
- Fitted by MLE on i.i.d. returns, the model mistakes volatile periods for frequent jumps
- Nonparametric tests relative to local volatility flag only a few candidate jumps per year in daily data

The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t [Heston \(1993\)](#)
 - ▶ $d \ln S_t = (\mu - \frac{1}{2} v_t) dt + \sqrt{v_t} dW_t^{(1)}$
 - ▶ $dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ ; ξ = volatility of volatility
 - ▶ Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero [Feller \(1951\)](#)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
 - ▶ Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ [Lord, Koekkoek & van Dijk \(2010\)](#); an alternative discretisation, the quadratic-exponential scheme [Andersen \(2008\)](#)

Heston Parameters from the VIX

- Proxy $y_t = (\text{VIX}_t/100)^2$ in place of v_t ; discretised CIR regression
 - ▶ $\frac{y_{t+\Delta t} - y_t}{\sqrt{y_t}} = \kappa\theta \frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
 - ▶ $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%; Feller ratio of the proxy $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t : it is a risk-neutral expectation of future variance, affine in v_t (next slide)
 - ▶ For real-world paths we set θ to the realised variance; $\hat{\xi}$ is a lower bound for ξ

What the VIX Identifies in Heston

□ Under Q : $y_t = \frac{1}{\tau} \mathbb{E}^Q \left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t \right] = a + b v_t$, $\tau = 30/365$ (Seminar A6)

▶ $b = \frac{1 - e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1 - b)$: affine in v_t , with Q -parameters

□ What the regression on y_t recovers

▶ κ : the P -speed, since an affine map keeps the dynamics of v_t ; ρ : unchanged, correlation is scale-free

▶ ξ : attenuated; the diffusion of y is $\xi \sqrt{b(y - a)}$, so $\xi_y = b \xi \sqrt{v/(a + bv)}$ varies with the state and equals $b \xi$ only at $v_t = \theta^Q$

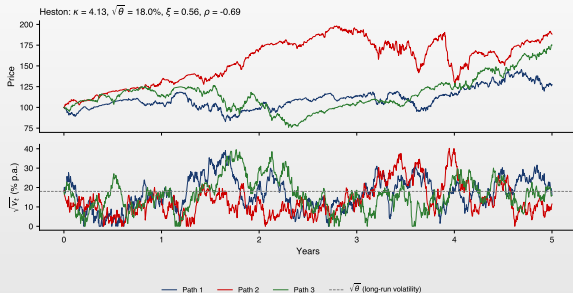
▶ The long-run level $a + b \theta^P$ mixes θ^Q and θ^P : recover $\theta^P = (\theta_y - a)/b$ before a Feller check on v_t

κ^Q	1	2,5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

□ $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$; proper estimation fits the shifted square-root dynamics of y , or v_t and the observation equation jointly

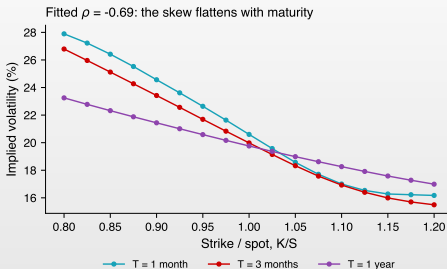
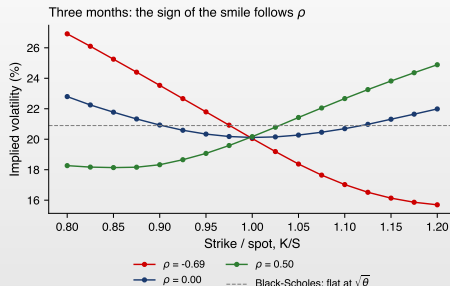
□ κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices [Chernov & Ghysels \(2000\)](#), or moments of realised variance [Bollerslev & Zhou \(2002\)](#)

Simulated Heston Paths



- Three five-year paths with the VIX-based $\hat{\kappa}$, $\hat{\xi}$ (uncorrected), $\hat{\rho}$ and θ equal to the realised S&P 500 variance; top: price, bottom: $\sqrt{V_t}$
- Clustering now comes from the model; price falls coincide with volatility spikes (simulated return–variance correlation -0.69 , the leverage effect through ρ)
- With θ at the realised variance the Feller ratio is $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the CIR variance can reach zero; negative values arise only in the Euler approximation, and full truncation sets them to zero (1.4% of the steps)

A Preview of Chapter 12: the Volatility Smile



- Black-Scholes implied volatilities of Heston call prices (Monte Carlo, 100,000 antithetic paths); left: effect of ρ at 3 months; right: maturities with the fitted ρ

Reading the Smile Chart

- ▣ **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price
 - ▶ Under GBM it is the same for every strike: a flat line
- ▣ With $\hat{\rho} = -0.69$, three months: 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%
 - ▶ A downward **skew**, as in equity-index options; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%)
- ▣ The skew flattens with maturity: from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year
- ▣ Illustration only: the VIX-regression estimates are used as Q -parameters; pricing needs κ^Q , θ^Q calibrated to option prices, as in Chapter 12

Affine Jump-Diffusions

- A model is **affine** if drift, variance and jump intensity are affine functions of the state X_t [Duffie, Pan & Singleton \(2000\)](#)
 - ▶ Vasicek, CIR, Merton, Heston and Bates are all affine
- Then the conditional characteristic function is exponential-affine in the state
 - ▶ $\mathbb{E} \left[e^{iu^\top X_\tau} \mid X_t \right] = \exp \left(\alpha(\tau) + \beta(\tau)^\top X_t \right)$, with α, β solving Riccati ordinary differential equations
- This is why Heston has “only the characteristic function” (Section 4): option prices follow by Fourier inversion
- The Vasicek bond of Section 8 is the simplest case: $A(\tau)$ and $B(\tau)$ solve Riccati equations

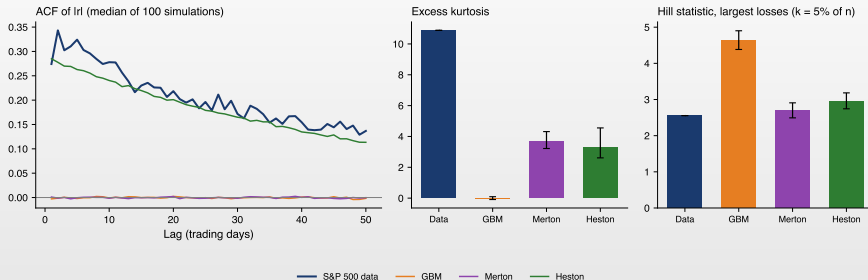
From GARCH to Continuous Time

- Nelson (1990): as the time step shrinks, GARCH(1,1) converges to a continuous-time stochastic-volatility diffusion
 - ▶ $dv_t = \kappa(\theta - v_t)dt + \xi v_t dW_t^{(2)}$: like Heston, but with v_t instead of $\sqrt{v_t}$
- The GARCH persistence $\alpha + \beta$ of Chapter 5 maps to $\kappa \approx (1 - \alpha - \beta)/\Delta t$
 - ▶ $\alpha + \beta = 0.99$ with daily data gives $\kappa \approx 2.5$ per year, a half-life of about 70 days
- Discrete-time GARCH for forecasting and risk; continuous-time stochastic volatility for pricing and hedging
- Caution: GARCH and its limit are **not** asymptotically equivalent experiments (one noise source against two): inference can differ even as $\Delta t \rightarrow 0$ Wang (2002)
- Further reading: Čížek, Härdle & Weron (2011) (Heston model chapter), Härdle, Chen & Overbeck (2017)

Recap: Stochastic Volatility

- ▣ Heston: variance is a mean-reverting CIR process correlated with returns
- ▣ Parameters can be read from the VIX: fast mean reversion, high volatility of volatility, strong leverage
- ▣ Generates volatility clustering and, through ρ , the implied-volatility skew
- ▣ GARCH is its discrete-time cousin

Which Model Reproduces the S&P 500?



- 100 simulated samples of 9 245 days from each fitted model: median ACF of $|r_t|$, excess kurtosis and the Hill statistic of the largest losses, with 90% ranges

Reading the Model Comparison

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	-0.36	0.00	-0.32	0.00
Hill statistic	2.56	4.64	2.70	2.95
ACF of $ r $, lag 1	0.27	0.00	0.00	0.28
ACF of $ r $, sum of lags 1–20	5.32	0.00	0.00	4.78

- Medians of 100 simulations; Hill statistic of the largest losses with $k = 5\%$ of all returns (lower = heavier tail): a tail index only under regular (Pareto-type) variation, descriptive otherwise; data: 2.97, 2.56, 2.04 for $k = 2.5\%$, 5% , 10%
- These fitted specifications: Merton gets the Hill statistic and skewness but no clustering; Heston, with the VIX-based parameters, gets the clustering but no skewness
- No fitted model reaches the kurtosis of 10.9; a Heston model with a lower Feller ratio could (short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$), but the evidence of [Andersen, Benzoni & Lund \(2002\)](#) and [Eraker, Johannes & Polson \(2003\)](#) favours jumps and stochastic volatility (SVJ, [Bates \(1996\)](#))

Which Model When?

Model	Captures	Typical use
Geometric Brownian motion	Positive prices, $\sqrt{t}$ scaling	Black–Scholes benchmark, long-horizon planning
Ornstein–Uhlenbeck, CIR	Mean reversion	Interest rates, spreads, volatility indices
Merton jump-diffusion	Gaps, short-horizon fat tails	Short-dated options, credit events, crypto
Heston stochastic volatility	Clustering, leverage, skew	Equity-index options, volatility trading
Bates (Heston + jumps)	All of the above	Option surfaces across maturities

Beyond This Chapter

- ▣ Heavier tails: Lévy processes and α -stable distributions [Pele \(2014\)](#)
 - ▶ Crypto-assets form a distinct class by the shape of their return distributions [Pele et al. \(2023\)](#)
- ▣ Rough volatility: volatility paths rougher than Brownian motion [Gatheral, Jaisson & Rosenbaum \(2018\)](#)
 - ▶ High-frequency information about Bitcoin volatility and tail risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- ▣ Exercises with solutions on all models of this chapter: [Borak, Härdle & López-Cabrera \(2013\)](#)
- ▣ Next, Chapter 12: pricing options by replication and reading the volatility surface

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $dX_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$; the answer depends on the sampling frequency and on the model
 - ▶ Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance; the 5-minute bipower test (Chapter 9): 2.8% for SPY
 - ▶ Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
 - ▶ Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t , look like jumps in coarse data
- Formal question: $H_0: \lambda = 0$ (continuous path with stochastic volatility) against $H_1: \lambda > 0$
 - ▶ Non-standard: $\lambda = 0$ lies on the boundary and (μ_J, σ_J) are unidentified under H_0 , so the LR statistic is not χ^2

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate estimates by frequency, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.
- 2. **Competing hypotheses**, with distinct testable implications
 - ▶ For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.
- 3. **Data and code**: replicate the chapter's estimate first
 - ▶ Write Python that reads `data/market/GSPC.INDX.csv` and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points). Target: $\hat{\lambda} = 102$ per year, jump share 71%

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at $t-1$; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.
- 5. **Critique**: ask the assistant to attack the inference
 - ▶ Act as a referee: why is the LR test of $\lambda = 0$ non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?
- 6. **Report**: all frequencies and filters, bootstrap critical values, code and the AI log

What the Researcher Must Check

- **Boundary and unidentified parameters:** the LR for $\lambda = 0$ has no χ^2 limit; test: parametric bootstrap under H_0
- **Look-ahead in the filter:** a two-sided (smoothed) volatility absorbs jumps artificially; test: only $\mathcal{F}_{t-1}$ in the standardisation
- **Frequency mismatch:** daily and 5-minute jump shares answer different questions; test: report the share at every frequency on a common sample
- **Multimodal likelihood:** a Poisson mixture has local optima; test: many starting points and a profile likelihood in λ
- **Discretisation:** Euler steps on a coarse grid create spurious large moves; test: exact simulation where it exists
- **Invented numbers:** claims such as “jumps explain 10% of variance” need a table and a DOI; check both

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- Replication: Merton on S&P 500 daily returns, $\hat{\lambda} = 102$ per year, jump share 71% (Section 9)
- A typical assistant proposal: “71% of variance comes from jumps”. Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility ($\lambda_{\text{EWMA}} = 0.94$, known at $t - 1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (s.e.)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a “jump” about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share; Lee–Mykland finds 0.87 jumps per year and 5-minute data give 2.8%. Code: `Quantlets/Ch_11/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: how much of S&P 500 and Bitcoin variance is discontinuous, at which frequency, and are the detected “jumps” drift bursts?
- Replicate first: Merton jump share 71% (daily S&P 500), Lee–Mykland 0.87 jumps per year, bipower jump share 2.8% (SPY, 5 minutes, Chapter 9)
- Deliverables
 - ▶ A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run
 - ▶ The jump share at 5, 15, 30 minutes and daily; the drift-burst statistic of Christensen, Oomen & Renò (2022) around April 2025; a bootstrap LR test of Heston (null, no jumps) against Bates
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI_USE.md and every error the assistant made in AI_ERRORS.md: github.com/danpele/MFM/tree/main/project

Summary

- Brownian motion is the universal limit of random walks; its quadratic variation t drives Itô calculus
- Itô's lemma gives explicit solutions (GBM, Ornstein–Uhlenbeck, Merton); simulate exactly or with Euler–Maruyama and Milstein
- Identification: infill pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased
- Girsanov changes only the drift; Feynman–Kac turns a Q -expectation into a PDE; the VIX is a Q -object, affine in v_t
- Data: in our fits jumps help with the tails and stochastic volatility with the clustering; the literature favours models with both

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Applied Computational Finance
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

References (1/5)

- ▣ Aït-Sahalia, Y. (1996). Testing continuous-time models of the spot interest rate. *Review of Financial Studies*, 9(2), 385–426.
- ▣ Aït-Sahalia, Y. (2002). Maximum likelihood estimation of discretely sampled diffusions: a closed-form approximation approach. *Econometrica*, 70(1), 223–262.
- ▣ Andersen, L. (2008). Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3), 1–42.
- ▣ Andersen, T.G., Benzoni, L., Lund, J. (2002). An empirical investigation of continuous-time equity return models. *Journal of Finance*, 57(3), 1239–1284.
- ▣ Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- ▣ Bandi, F.M., Phillips, P.C.B. (2003). Fully nonparametric estimation of scalar diffusion models. *Econometrica*, 71(1), 241–283.
- ▣ Bates, D.S. (1996). Jumps and stochastic volatility: exchange rate processes implicit in Deutsche Mark options. *Review of Financial Studies*, 9(1), 69–107.
- ▣ Billingsley, P. (1999). *Convergence of Probability Measures*, 2nd ed. Wiley.
- ▣ Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- ▣ Bollerslev, T., Zhou, H. (2002). Estimating stochastic volatility diffusion using conditional moments of integrated volatility. *Journal of Econometrics*, 109(1), 33–65.
- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Chan, K.C., Karolyi, G.A., Longstaff, F.A., Sanders, A.B. (1992). An empirical comparison of alternative models of the short-term interest rate. *Journal of Finance*, 47(3), 1209–1227.
- ▣ Chernov, M., Ghysels, E. (2000). A study towards a unified approach to the joint estimation of objective and risk neutral measures for the purpose of options valuation. *Journal of Financial Economics*, 56(3), 407–458.

References (2/5)

- ▣ Christensen, K., Oomen, R.C.A., Podolskij, M. (2014). Fact or friction: jumps at ultra high frequency. *Journal of Financial Economics*, 114(3), 576–599.
- ▣ Christensen, K., Oomen, R.C.A., Renò, R. (2022). The drift burst hypothesis. *Journal of Econometrics*, 227(2), 461–497.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). *Statistical Tools for Finance and Insurance*, 2nd ed. Springer.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- ▣ Duffie, D., Pan, J., Singleton, K. (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, 68(6), 1343–1376.
- ▣ Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8), 549–560.
- ▣ Eraker, B., Johannes, M., Polson, N. (2003). The impact of jumps in volatility and returns. *Journal of Finance*, 58(3), 1269–1300.
- ▣ Feller, W. (1951). Two singular diffusion problems. *Annals of Mathematics*, 54(1), 173–182.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- ▣ Girsanov, I.V. (1960). On transforming a certain class of stochastic processes by absolutely continuous substitution of measures. *Theory of Probability and Its Applications*, 5(3), 285–301.
- ▣ Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.

References (3/5)

- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Heston, S.L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options*. *Review of Financial Studies*, 6(2), 327–343.
- ▣ Higham, D.J. (2001). *An algorithmic introduction to numerical simulation of stochastic differential equations*. *SIAM Review*, 43(3), 525–546.
- ▣ Honoré, P. (1998). *Pitfalls in estimating jump-diffusion models*. Working paper, SSRN.
- ▣ Itô, K. (1944). *Stochastic integral*. *Proceedings of the Imperial Academy*, 20(8), 519–524.
- ▣ Kac, M. (1949). *On distributions of certain Wiener functionals*. *Transactions of the American Mathematical Society*, 65(1), 1–13.
- ▣ Kloeden, P.E., Platen, E. (1992). *Numerical Solution of Stochastic Differential Equations*. Springer.
- ▣ Lee, S.S., Mykland, P.A. (2008). *Jumps in financial markets: a new nonparametric test and jump dynamics*. *Review of Financial Studies*, 21(6), 2535–2563.
- ▣ Lord, R., Koekkoek, R., van Dijk, D. (2010). *A comparison of biased simulation schemes for stochastic volatility models*. *Quantitative Finance*, 10(2), 177–194.
- ▣ Merton, R.C. (1973). *Theory of rational option pricing*. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- ▣ Merton, R.C. (1976). *Option pricing when underlying stock returns are discontinuous*. *Journal of Financial Economics*, 3(1–2), 125–144.
- ▣ Milstein, G.N. (1975). *Approximate integration of stochastic differential equations*. *Theory of Probability and Its Applications*, 19(3), 557–562.
- ▣ Nelson, D.B. (1990). *ARCH models as diffusion approximations*. *Journal of Econometrics*, 45(1–2), 7–38.

References (4/5)

- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Osborne, M.F.M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- ▣ Pele, D.T. (2014). A SAS approach for estimating the parameters of an alpha-stable distribution. *Procedia Economics and Finance*, 10, 68–77.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pritsker, M. (1998). Nonparametric density estimation and tests of continuous time interest rate models. *Review of Financial Studies*, 11(3), 449–487.
- ▣ Samuelson, P.A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–39; reprinted in *Henry P. McKean Jr. Selecta* (2015), Birkhäuser.
- ▣ Shreve, S.E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer.
- ▣ Stanton, R. (1997). A nonparametric model of term structure dynamics and the market price of interest rate risk. *Journal of Finance*, 52(5), 1973–2002.
- ▣ Tang, C.Y., Chen, S.X. (2009). Parameter estimation and bias correction for diffusion processes. *Journal of Econometrics*, 149(1), 65–81.
- ▣ Uhlenbeck, G.E., Ornstein, L.S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5), 823–841.
- ▣ Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.
- ▣ Wang, Y. (2002). Asymptotic nonequivalence of GARCH models and diffusions. *Annals of Statistics*, 30(3), 754–783.

References (5/5)

- Wiener, N. (1923). [Differential-space](#). *Journal of Mathematics and Physics*, 2(1–4), 131–174.
- Yu, J. (2012). [Bias in the estimation of the mean reversion parameter in continuous time models](#). *Journal of Econometrics*, 169(1), 114–122.